Statistics of Financial Markets

Chapter 14: Review

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Outline

Course Summary

- Overview of all topics covered in Chapters 1–13
- Efficient Market Hypothesis and random walk models
- Variance ratio tests and stylized facts of financial returns
- Volatility modeling: ARCH, GARCH, and asymmetric extensions
- Risk measures: VaR and Expected Shortfall
- Credit scoring and machine learning in finance
- Cryptocurrency markets and stablecoins
- Systemic risk measurement and network analysis

Key Concepts

- Market efficiency and its three forms
- Random walk types (RW1, RW2, RW3) and their tests
- Stylized facts: heavy tails, volatility clustering, leverage effect
- GARCH family: specification, estimation, model selection
- VaR estimation methods and backtesting
- Logistic regression and ROC/AUC for scoring models
- LASSO, Random Forest, and model validation
- CoVaR, FRM, and systemic risk networks

Exam Preparation

- Exam format and structure
- Key definitions and theorems to review
- Practice problems and worked examples
- Tips for computational questions (Python/R)
- Frequently asked questions from previous years

References

- Franke, J., Härdle, W.K., Hafner, C.M. (2019). Statistics of Financial Markets: An Introduction.
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- McNeil, A.J., Frey, R., Embrechts, P. (2015). Quantitative Risk Management.
- Tsay, R.S. (2010). Analysis of Financial Time Series.